

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456299.74195 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.053 +/- 0.025
Transit depth (Rp/Rs)^2	0.28 +/- 0.27 [%]
Orbital Inclination (inc)	87.64 +/- 1.88
Airmass coefficient 1 (a1)	0.9717 +/- 0.0081
Airmass coefficient 2 (a2)	0.0185 +/- 0.0088
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	#2 - [83, 134]
Optimal Aperture	4.26
Optimal Annulus	7.46
Transit Duration (day)	0.0917 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.053 ± 0.025 vs 0.1241 (deviation 2.76σ)
Duration fit vs published	0.0917 d vs 0.1317 d (deviation 4.16σ)
Mid-time O-C	1.5 min
Depth SNR	2.1
Residual scatter	0.83 %

Light curve

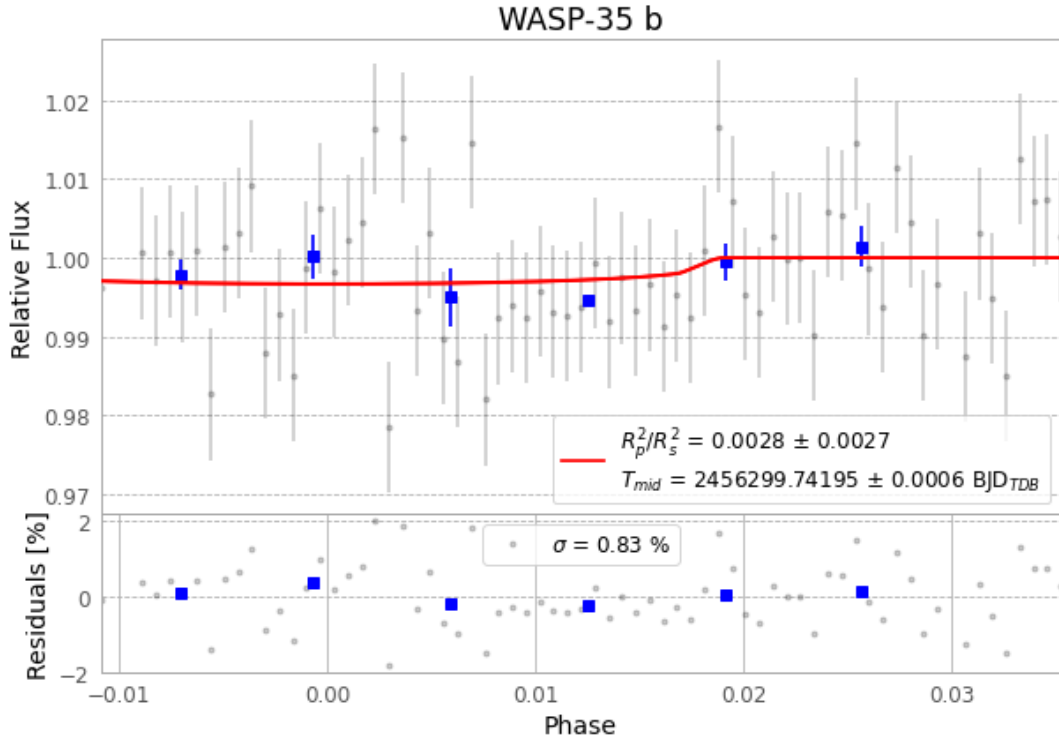


Figure 1 — FinalLightCurve_WASP-35 b_2013-01-06.png (SHA-256 6e5d13afb6805d17...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 303], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1426a94dadf08bea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-35130107045113.FITS → WASP-35130107082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-130107.log (e3b77fb3bf6a...), FinalLightCurve_WASP-35 b_2013-01-06.png (6e5d13afb680...), FinalLightCurve_WASP-35 b_2013-01-06.csv (5fff0ef1aab3...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:40Z.